DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2024

Course: B. Tech. **Branch: Common to all branches** **Semester: I**

Subject Code & Name: BTES105 - Energy and Environment Engineering

Max Marks: 60 **Date: 09/07/2024** **Duration: 3 Hr.**

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		**12**
A)	**Difference between :- Thermo electric and Thermionic generators**	**CO1**	**6**
B)	**Short Notes:-** **i) Conventional Energy Sources** **ii) Steam Power Generation**	**CO2**	**6**
C)	**Explain in detail Gas Turbine Power Plant with neat Diagram.**	**CO3**	**6**
Q.2	**Solve Any Two of the following.**		**12**
A)	**Explain in detail Magneto Hydro Dynamics (MHD) with neat diagram.**	**CO3**	**6**
B)	**Explain the Fuel Cell with neat diagram.**	**CO2**	**6**
C)	**Short Notes:-** **i) Biogas** **ii) Biomass**	**CO1**	**6**
Q. 3	**Solve Any Two of the following.**		**12**
A)	**Write the Energy conservation in electric furnaces, ovens and boilers lighting techniques.**	**CO1**	**6**
B)	**Write the Methods and techniques of energy conservation in ventilation and air conditioners, compressors, pumps, fans and blowers.**	**CO2**	**6**
C)	**Short Notes:-** **i) Maximum energy efficiency** **ii) Maximum cost effectiveness.**	**CO3**	**6**

Q.4	**Solve Any Two of the following.**		**12**
A)	**Write in detail the control measures in Air pollution.**	**CO3**	**6**
B)	**Write in detail about Air pollution Sources and Effects.**	**CO4**	**6**
C)	**Short Notes:-** **i) Particulate Emission** **ii) Air quality standards**	**CO5**	**6**
Q. 5	**Solve Any Two of the following.**		**12**
A)	**Write the Water Pollution Effects and Control Measures.**	**CO3**	**6**
B)	**Short Notes :-** **i) Thermal Pollution** **ii) Soil Pollution**	**CO4**	**6**
C)	**Write the Noise pollution effects and control measures.**	**CO5**	**6**

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